

Was She A Witch?

A Data-Driven Approach to Medieval Cancel Culture

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1. Introduction

The Scottish witch-hunts (c. 1563–1736) represent one of the most intense periods of judicial prosecution in European history. Unlike many continental neighbors, Scotland's prosecutions were characterised by a highly centralised legal system and a unique theological obsession with the demonic pact. While the popular imagination often views these trials as arbitrary outbursts of mass hysteria, historical scholarship suggests that accusations were often driven by specific, measurable socio-economic tensions. As noted by the article *They weren't witches; they were women: The witch-hunts and their repercussions* (2024), "Accusations of witchcraft were often based on unfounded rumours, superstitions, and even personal grudges. Individuals accused of witchcraft were frequently those who deviated from societal norms or challenged traditional gender roles".

In Scotland, the legal definition of witchcraft shifted significantly following the Reformation. Under the Witchcraft Act of 1563, the essence of the crime was no longer just *maleficium* (harmful magic) but the pact with the Devil. This shift turned a private dispute between neighbours into a capital crime against the state and God. Quantifying the probability of an individual being specifically accused of a "pact" requires an understanding of how their social identity intersected with these theological and legal categories.

The *Survey of Scottish Witchcraft* provides a granular dataset to test whether specific characteristics acted as significant predictors for being targeted by the authorities. By applying a Bayesian Logistic Regression to the Scottish data, we aim to move beyond anecdotal evidence: by treating the demonic pact accusation as a binary outcome, we can evaluate which socio-economic profiles were statistically most "at risk" of being categorised as an enemy of the state.

Can we predict the probability of an individual being accused of a demonic pact based on their socio-economic characteristics?

We anticipate that if this is the case, we should observe:

- **P1:** Being female should increase the probability of accusation.
- **P2:** Being single should increase the probability of accusation.

- **P3:** Being in an irregular union should increase the probability of accusation relative to socially normative marital situations.
- **P4:** The probability of accusation should increase as age increases.

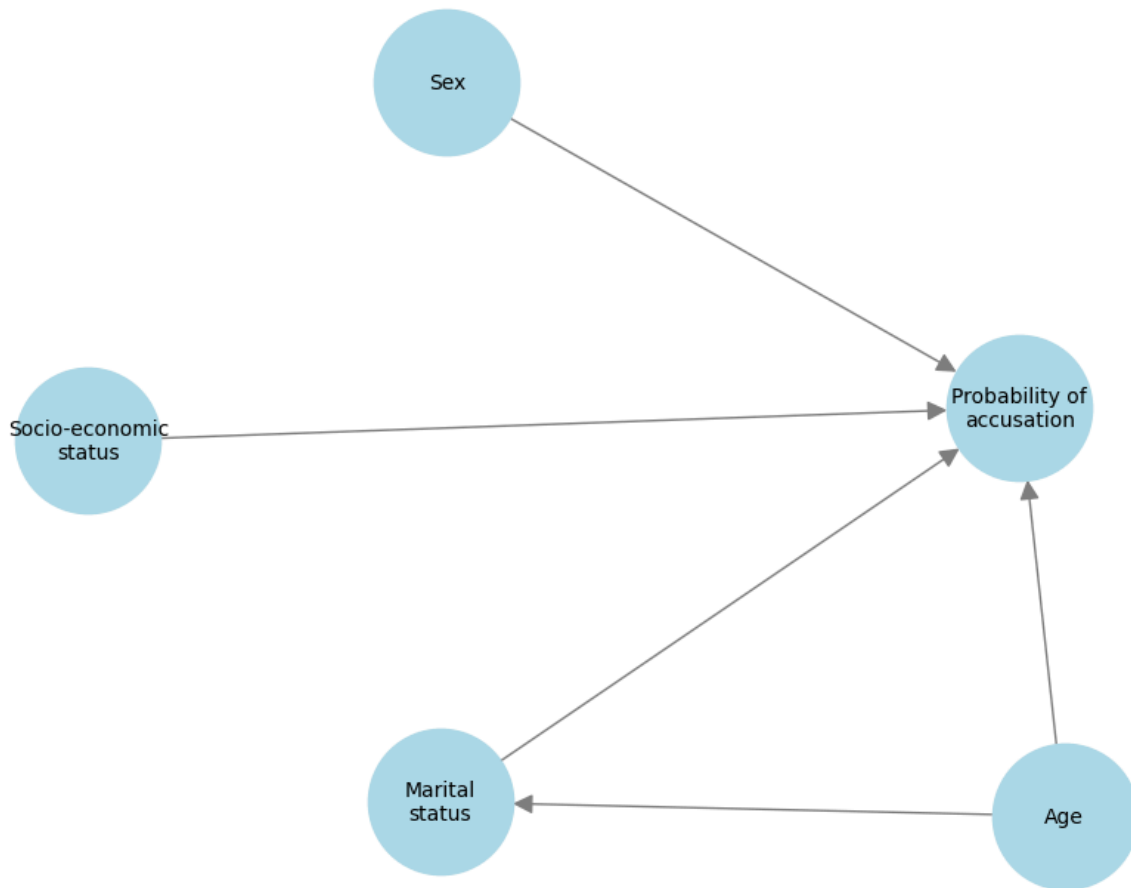


Figure 1: Directed acyclic graph representing the hypothesised causal relationships between socio-economic characteristics and the probability of being accused of a demonic pact

2. Data

a. Dataset description

The dataset utilised in this study was sourced via Kaggle, where it was republished by the user The Devastator in 2024. The underlying data is entirely derived from *The Survey of Scottish Witchcraft*, a comprehensive academic project funded by the Economic and Social Research Council (ESRC). The original survey was directed by Dr. Julian Goodare (University of Edinburgh) and co-directed by Dr. Louise Yeoman, with core research and database development conducted by Dr. Joyce Miller and Lauren Martin.

The database synthesises and expands upon three foundational historical surveys:

- *Calendar of Cases of Witchcraft* (1938), compiled by George Fraser Black, covering Scottish cases from 1510 to 1727.
- *Source-Book of Scottish Witchcraft* (1977), a comprehensive compilation funded by the Social Science Research Council.
- *Scottish Witch Hunt Data Base* (1990s), a digital CD-ROM project that expanded directly upon the Source-Book.

The final dataset contains records for 3,837 accused individuals, extracted from primary contemporary court documents spanning the enforcement of the Scottish Witchcraft Act from 1563 until its repeal in 1736. Crucially for this analysis, the database compiles detailed socio-demographic and legal characteristics for each individual, including sex, age, marital status, socio-economic standing, and the specific legal motives or characteristics of the trial.

b. Data description

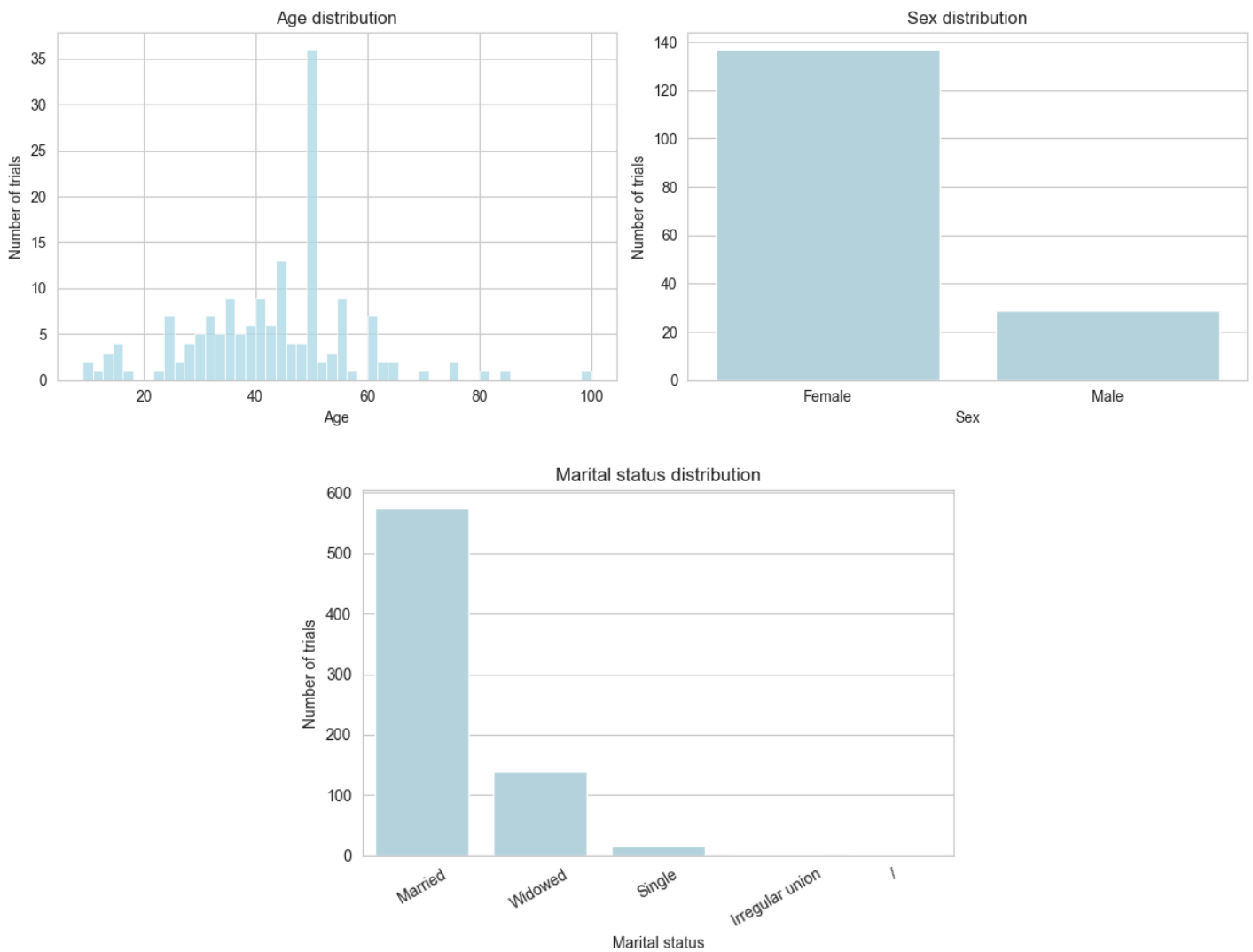


Figure 2: Distributions of socio-economic variables in our dataset

3. Methods

a. Model

To test our hypotheses, we performed a Bayesian logistic regression using the PyMC library. Within this probabilistic framework, the posterior probability distribution of each regression parameter, conditioned on the observed data, is estimated using a Markov Chain Monte Carlo (MCMC) sampling algorithm. The model is defined by the following formulation

$$\text{Likelihood: } y_i \sim \text{Bernoulli}(p_i)$$

$$\text{Link Function: } \text{logit}(p_i) = \alpha + \sum_{j=1}^K \beta_j X_{i,j}$$

$$\text{Priors: } \alpha \sim \text{Normal}(0, 1)$$

$$\beta_j \sim \text{Normal}(0, 1), \quad \forall j \in \{1, \dots, K\}$$

where:

- $y_i \in \{0, 1\}$ is the observed outcome for individual i (1 for people accused of a demonic pact, 0 for other accusation);
- p_i is the probability that individual i is accused of a demonic pact;
- α represents the intercept;
- β_j represents the coefficient for feature j ;
- $X_{i,j}$ is the value of feature j for individual i ;
- K is the total number of features ($n_features$), which corresponds to the predictors (sex, age, marital status, including dummy variables).

b. Bayesian logistic regression

i. Statistical tool presentation

Logistic regression is the standard statistical approach for modeling a binary outcome $y \in \{0, 1\}$. The frequentist approach (MLE) seeks a single point estimate for the parameters.

In contrast, the Bayesian approach treats parameters as probability distributions. The model is built on three pillars: the Likelihood, the Link Function, and the Prior.

- **Likelihood:** Since each observation is a "success" (accused) or "failure" (not accused), we use the Bernoulli distribution to model the data generating process:

$$P(y|\alpha, \beta, X) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

- **Link Function (Logit):** We model the probability p_i of an accusation using the logistic (sigmoid) function. This function transforms a linear combination of predictors (such as age, social status, or gender) into a probability between 0 and 1:

$$p_i = \sigma(\alpha + \beta^T x_i) = \frac{1}{1 + e^{-(\alpha + \beta^T x_i)}}$$

Where α is the intercept (baseline risk) and β is the vector of coefficients (the weight of each predictor).

- **Prior Distributions:** Priors represent our knowledge before observing the data. In our model, we utilise Gaussian priors:

$$\beta \sim \mathcal{N}(0, 1) \quad \text{and} \quad \alpha \sim \mathcal{N}(0, 1)$$

By centering these at zero with a standard deviation of 1, we apply regularisation. This expresses a belief that predictors likely have small effects, which prevents the model from overfitting.

The goal of the analysis is to find the Posterior Distribution, which is the probability of the parameters given the observed data. This is defined by Bayes' Theorem:

$$P(\alpha, \beta | y, X) \propto \text{Likelihood} \times \text{Prior}$$

Because the denominator of Bayes' Theorem (the marginal likelihood) is mathematically impossible to calculate for logistic regression, we use Markov Chain Monte Carlo (MCMC). We also use the NUTS algorithm (No-U-Turn Sampler) to draw samples and map out the shape of the posterior distribution. This allows us to calculate the credible intervals of 94%.

ii. In-Depth Explanation

Our approach is justified by the alignment between the mathematical framework of our analysis and the constraints of the historical data: 16th and 17th-century legal records are structurally flawed, suffering from survival bias, missing data, and regional inconsistencies. While the logistic regression maintains a relative interpretability of the results, the bayesian framework allows an honest quantification of uncertainty, providing a mathematically transparent reflection of the limits of the data.

iii. History

Early applications of logistic regression typically relied on subject-matter expertise for variable selection rather than automated Bayesian methods. While the theoretical foundations

for Bayesian inference existed, they were rarely used for logistic regression because the posterior distribution does not have a closed-form solution.

The field shifted dramatically in the early 1990s with the widespread adoption of Markov Chain Monte Carlo (MCMC) algorithms, such as Gibbs sampling and the Metropolis-Hastings algorithm. These simulation-based methods allowed statisticians to draw samples from complex posterior distributions that were previously impossible to calculate. This era enabled the application of Bayesian logistic regression to diverse fields like medicine and social sciences.

As datasets grew, researchers developed specialised Bayesian models to handle thousands of predictors. Significant work emerged in sparse Bayesian models, which use hierarchical priors to “shrink” irrelevant coefficients to zero, effectively performing feature selection and estimation simultaneously. This was particularly impactful in fields like text categorisation.

In the current era, traditional MCMC can be too slow for millions of data points. Modern research focuses on Variational Bayesian (VB) methods and Polynomial Approximations (PASS-GLM). These techniques treat inference as an optimisation problem, providing significant speedups while maintaining the benefits of uncertainty quantification. Recent state-of-the-art models, such as BRIDGE (2026), demonstrate how Bayesian logistic regression is now integrated with hierarchical structures to provide interpretable, clinically relevant predictions that rival complex machine learning algorithms like XGBoost.

4. Results

The posterior estimates obtained from the Bayesian logistic regression (see *Figure 3*) suggest that most socio-economic variables have relatively uncertain effects on the probability of being accused of a demonic pact.

Regarding sex, the posterior mean associated with the variable *Sex_Male* is very close to zero ($\beta = -0.006$), and its 94% HDI largely overlaps zero. This indicates that the observed data do not provide substantial posterior evidence for a meaningful difference in accusation probability between men and women. Consequently, prediction **P1**, stating that women were more likely to be accused, **is not strongly supported by the model**.

Age appears to have a slightly negative posterior effect ($\beta = -0.247$), suggesting that older individuals may have been somewhat less likely to be accused. However, the corresponding HDI still includes zero, which indicates substantial posterior uncertainty. Therefore, the data only weakly support the existence of an age effect and **does not support prediction P4**.

Marital status produces more interesting posterior patterns. The coefficient associated with the category *MaritalStatus_Single* is positive ($\beta = 0.327$), meaning that single individuals may have had a higher posterior probability of accusation compared to the reference category. Although the 94% HDI still overlaps zero, the posterior distribution leans toward a positive effect, providing **moderate but inconclusive support for prediction P2**.

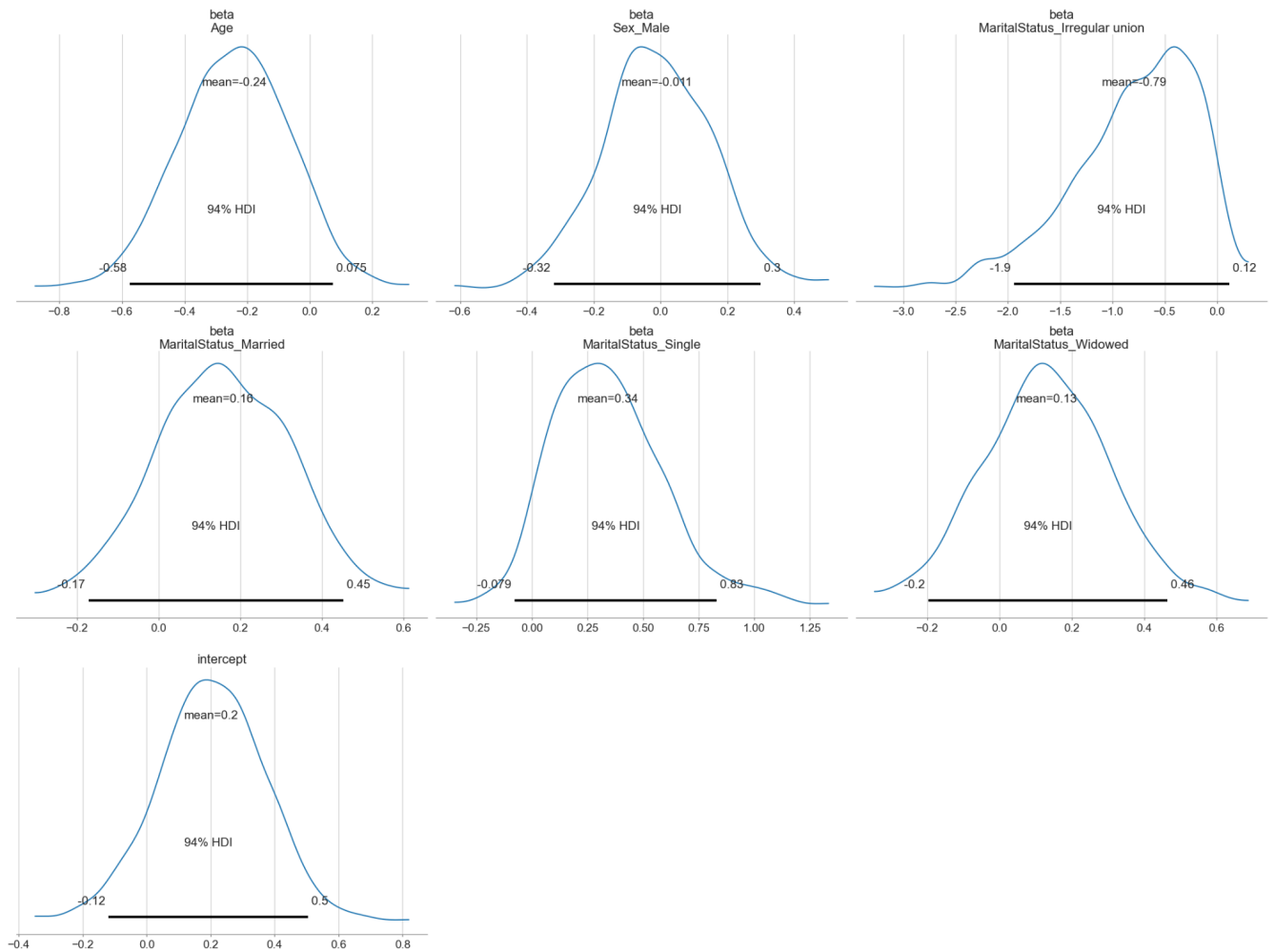


Figure 3: Posterior distributions of socio-economic characteristics on demonic pact accusations

The category *MaritalStatus_Irregular union* shows the strongest effect in magnitude ($\beta = -0.758$), suggesting a potentially lower probability of accusation relative to the baseline category. Nevertheless, uncertainty remains important, as the HDI still includes zero. The categories *MaritalStatus_Married* and *MaritalStatus_Widowed* both exhibit posterior distributions centered close to zero, indicating **little evidence for a substantial influence on accusation probability (P3)**.

Overall, the posterior distributions suggest that while some socio-economic characteristics may have influenced accusations, **the available data do not provide strong posterior certainty for most effects included in the model**.

5. Conclusion

a. Global conclusion

This Bayesian analysis aimed to estimate whether socio-economic characteristics influenced the probability of being accused of entering a demonic pact during Scottish witch trials. The

results indicate that most estimated effects remain uncertain, with posterior credible intervals frequently overlapping zero. From a Bayesian perspective, this does not imply that the null hypothesis is “proven”; it rather means that the observed data are compatible with both the presence and the absence of substantial effects for several variables. The analysis therefore highlights the importance of uncertainty quantification when studying historical social phenomena.

Among the variables included in the model, marital status appears to produce the strongest posterior tendencies, particularly for single individuals, whose posterior distribution suggests a moderately increased probability of accusation. In contrast, sex does not appear to exert a substantial effect in the present dataset, despite common historical assumptions regarding gender and witchcraft accusations.

b. Epistemologic conclusion

Our contribution strongly shaped the direction of the project, both intellectually and personally. Initially centered on beer and consumption patterns, the project shifted toward witchcraft accusations due to a lack of data, but also because we were particularly interested in the social and historical dimensions of persecution. As two feminist people, we were drawn to the historical treatment of marginalised individuals, especially women, and to the ways social biases may have influenced accusations during witch trials. We also enjoyed working with historical records and transforming archival data into a quantitative analysis. Besides the methodological interest of Bayesian modelling, we appreciated the project because it connected data science, history, and social interpretation in a way that felt both original and meaningful.

c. Limitations

Several limitations should be acknowledged. First, the historical records used in this study are incomplete and may suffer from survival bias, as not all witchcraft cases were documented or preserved. Second, the model only includes a limited number of socio-economic variables and therefore cannot account for other potentially important factors, such as local political tensions, religious conflicts, or regional judicial practices. Finally, our Bayesian logistic regression assumes a linear relationship between the predictors and the log-odds of accusation. As suggested by the age coefficient, some effects may in reality be non-linear and therefore not fully captured by the current model.

6. Bibliography

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7. Appendix

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
beta[Age]	-0.238	0.178	-0.574	0.097	0.005	0.007	1092.0	743.0	1.0
beta[Sex_Male]	-0.022	0.173	-0.325	0.322	0.005	0.006	1006.0	744.0	1.0
beta[MaritalStatus_Irregular union]	-0.772	0.613	-1.976	0.120	0.029	0.028	582.0	358.0	1.0
beta[MaritalStatus_Married]	0.155	0.164	-0.159	0.452	0.006	0.005	836.0	679.0	1.0
beta[MaritalStatus_Single]	0.341	0.259	-0.168	0.773	0.010	0.008	802.0	630.0	1.0
beta[MaritalStatus_Widowed]	0.119	0.164	-0.180	0.442	0.006	0.005	880.0	726.0	1.0
intercept	0.203	0.165	-0.089	0.521	0.006	0.004	734.0	748.0	1.0

Appendix A: Table of results of the logistic regression